Muhammad Raffay Ismat

Software Engineer · Workflow Automation · Backend Development

Email | Linkedin | Github | Portfolio | Pakistan · Open to Relocation (EU)

PROFILE

Final-year Computer Science student with ~1 year of experience building AI-powered applications and workflow automations. Proven ability to independently design, develop, and ship projects spanning RAG pipelines, LLM security, and voice agents. Seeking a junior role in AI/ML engineering, backend development, intelligent automation or AI agent development.

SKILLS

Languages: Python

AI / ML: LangChain, RAG, OpenAI API, Vector Embeddings, Prompt Engineering, NLP

Automation: n8n, zapier, make, REST API Integration, Webhook Orchestration, Multi-step Pipelines

Backend & Cloud: Docker, DigitalOcean, Netlify, Vercel, Railway

Databases: MySQL, PostgreSQL, Supabase

Version Control: Git/Github, Unix Commands

EXPERIENCE

AI Software Engineer · Kaplan *2025 – Present*

- Built production-style AI applications in Python and JavaScript, integrating OpenAI APIs, vector databases, and third-party REST services.
- Designed and automated multi-step workflows in n8n including webhook triggers, conditional logic, and external API orchestration.
- Managed full project lifecycle independently — requirements, architecture, development, testing, and deployment.

PROJECTS

Air Quality RAG Chatbot · Python · LangChain · Vector Embeddings · Kaggle *2024*

- Built a retrieval-augmented generation chatbot that answers natural-language queries about global air pollution using semantic search over an embedded Kaggle dataset.
- Designed the embedding pipeline, vector store ingestion, and LLM synthesis layer end-to-end; chatbot returns context-grounded answers with source traceability.

Prompt Injection Firewall · Python · NLP · LLM Security *2025*

- Trained a classification model acting as a security proxy between users and a primary LLM, detecting and blocking prompt injection attempts in real time.
- Evaluated against an adversarial prompt test suite; demonstrated measurable reduction in injection pass-through versus an unguarded baseline.

JobReach — EU Startup Outreach Automation · Python · Tkinter · OpenAI API · Gmail SMTP *2026*

- Built a full-stack desktop automation tool in Python/Tkinter that scrapes 50+ European tech startups, generates personalised cold emails via OpenAI (gpt-4o-mini), and delivers them through Gmail SMTP with randomised send delays to mimic human behaviour.
- Designed a multi-step pipeline: lead scraping → per-company AI prompt generation using real company “about” data → email preview → scheduled send; persists all state to JSON for resume-safe campaigns targeting EU Schengen-area companies.

EDUCATION

BSc Computer Science · University of Central Punjab *2022 – 2027 (Expected)*

- Relevant coursework: Algorithms & Data Structures, Computer Networks, Theory of Computation, Databases, Operating Systems.

CERTIFICATIONS

Responsive Web Design · freeCodeCamp · 2021 · Verify
Introduction to Back-End Development · Meta · 2022 · Verify
Programming in Python · Meta · 2022 · Verify
Version Control · Meta · 2022 · Verify
Introduction to Databases for Back-End Development · Meta · 2022 · Verify